

Chapter 25: Statistics

Overview

Statistics (Thống kê) is the science of collecting, organizing, analyzing, and interpreting data. Descriptive statistics focuses on summarizing and describing the main features of a dataset using numerical measures and visualizations.

What you'll master:

- **Measures of Central Tendency:** Mean, median, mode, and quartiles
- **Measures of Dispersion:** Range, interquartile range, variance, and standard deviation
- **Data Visualization:** Interpreting histograms, box plots, and dot plots
- **Outlier Detection:** Identifying unusual values in datasets
- **Data Interpretation:** Drawing meaningful conclusions from statistical summaries

Real-world connections:

- Educational assessment and student performance analysis
- Quality control in manufacturing
- Public health and epidemiology
- Economic indicators and market research
- Scientific research and experimental design

Vocabulary & Terminology

Core Terms

English	IPA	Vietnamese	Definition
Statistics	/stəˈtɪstɪks/	Thống kê	The science of data collection and analysis
Data	/'dɜ:tə/	Dữ liệu	Information collected for analysis
Dataset	/'dɜ:təset/	Mẫu số liệu	A collection of data values

Population	/ˌpɒpjʊˈleɪʃən/	Tổng thể	The entire group being studied
Sample	/ˈsɑːmpəl/	Mẫu	A subset of the population
Variable	/ˈveəriəbəl/	Biến số	A characteristic that can vary
Frequency	/ˈfriːkwənsi/	Tần số	How often a value occurs
Distribution	/ˌdɪstriˈbjʊːʃən/	Phân phối	How data values are spread

Measures of Central Tendency

English	IPA	Vietnamese	Definition
Mean	/mi:n/	Số trung bình	The arithmetic average
Median	/ˈmi:diən/	Trung vị	The middle value when data is ordered
Mode	/məʊd/	Mốt	The most frequently occurring value
Quartile	/ˈkwɔ:təl/	Tứ phân vị	Values dividing data into four equal parts
First quartile (Q₁)	/fɜ:st ˈkwɔ:təl/	Tứ phân vị thứ nhất	The 25th percentile
Second quartile (Q₂)	/ˈsekənd ˈkwɔ:təl/	Tứ phân vị thứ hai	The median (50th percentile)
Third quartile (Q₃)	/θɜ:d ˈkwɔ:təl/	Tứ phân vị thứ ba	The 75th percentile

Measures of Dispersion

English	IPA	Vietnamese	Definition
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Range	/reɪndʒ/	Khoảng biến thiên	Difference between max and min
Interquartile range (IQR)	/ˌɪntəˈkwɔːtaɪl reɪndʒ/	Khoảng tứ phân vị	$Q_3 - Q_1$
Variance	/ˈveəriəns/	Phương sai	Average of squared deviations
Standard deviation	/ˈstændəd ˌdiːviˈeɪʃən/	Độ lệch chuẩn	Square root of variance
Deviation	/ˌdiːviˈeɪʃən/	Độ lệch	Distance from the mean
Outlier	/ˈaʊtlaɪər/	Giá trị bất thường	An unusually extreme value

Data Visualization

English	IPA	Vietnamese	Definition
Histogram	/ˈhɪstəgræm/	Biểu đồ tần số	Bar chart showing frequency distribution
Box plot	/bɒks plɒt/	Biểu đồ hộp	Visual summary using quartiles
Dot plot	/dɒt plɒt/	Biểu đồ chấm	Each data point shown as a dot
Stem-and-leaf plot	/stem ənd liːf plɒt/	Biểu đồ thân lá	Displays data preserving original values

 Theory

1. Measures of Central Tendency (Số đặc trưng đo xu thế trung tâm)

These measures describe the "center" or "typical value" of a dataset.

1.1 Mean (Số trung bình)

Definition: The **mean** (or arithmetic average) is the sum of all values divided by the number of values.

Formula: For dataset

$x_1, x_2, \dots, x_n$

:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Properties:

- Uses all data values
- Sensitive to outliers (extreme values can distort the mean)
- Unique (every dataset has exactly one mean)

When to use: When data is roughly symmetric without extreme outliers.

1.2 Median (Trung vị)

Definition: The **median** is the middle value when data is arranged in order.

Finding the median:

1. **Arrange data in ascending order**
2. **If n is odd:** Median = middle value (position $\frac{n+1}{2}$)
3. **If n is even:** Median = average of two middle values (positions $\frac{n}{2}$ and $\frac{n}{2} + 1$)

Properties:

- Not affected by outliers (robust measure)
- Divides data into two equal halves
- Better than mean for skewed distributions

When to use: When data contains outliers or is skewed.

1.3 Mode (Mốt)

Definition: The **mode** is the value that occurs most frequently.

Properties:

- A dataset can have no mode (all values occur once)
- A dataset can have multiple modes (bimodal, multimodal)
- Most useful for categorical data

When to use: For categorical data or to identify the most common value.

1.4 Quartiles (Tứ phân vị)

Definition: **Quartiles** divide ordered data into four equal parts.

- **Q₁** (first quartile): 25% of data is below this value
- **Q₂** (second quartile): The median (50% below)
- **Q₃** (third quartile): 75% of data is below this value

Finding quartiles:

1. **Find Q₂** (the median of the entire dataset)
2. **Find Q₁**: Median of the lower half (data below Q₂)
3. **Find Q₃**: Median of the upper half (data above Q₂)

Note: When n is odd, exclude the median when finding Q₁ and Q₃.

2. Measures of Dispersion (Số đặc trưng đo độ phân tán)

These measures describe how spread out the data is.

2.1 Range (Khoảng biến thiên)

Definition: The **range** is the difference between the maximum and minimum values.

$$\text{Range} = x_{\max} - x_{\min}$$

Properties:

- Simple to compute
- Uses only two values (ignores all others)
- Very sensitive to outliers

2.2 Interquartile Range (Khoảng tứ phân vị)

Definition: The **interquartile range (IQR)** is the range of the middle 50% of data.

$$\text{IQR} = Q_3 - Q_1$$

Properties:

- Robust to outliers (uses middle 50% only)
- Measures spread of the central data
- Used in outlier detection

2.3 Variance (Phương sai)

Definition: The **variance** measures the average squared deviation from the mean.

Population variance:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample variance (alternative formula):

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Interpretation:

- Larger variance $\rightarrow$ more spread out data
- Units are squared (e.g., if data is in cm, variance is in cm^2)

2.4 Standard Deviation (Độ lệch chuẩn)

Definition: The **standard deviation** is the square root of the variance.

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Properties:

- Same units as the original data
- Most commonly used measure of spread

- Approximately 68% of data lies within 1 standard deviation of the mean (for normal distributions)

When to use: When you need a measure of spread in the same units as the data.

3. Box Plots and Outlier Detection (Biểu đồ hộp và phát hiện giá trị bất thường)

3.1 Box Plot Structure

A **box plot** (box-and-whisker plot) visually displays the five-number summary:



Five-number summary: Min, Q₁, Q₂ (median), Q₃, Max

3.2 Outlier Detection

Definition: A value is considered an **outlier** if:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

Lower fence:

$$Q_1 - 1.5 \times \text{IQR}$$

Upper fence:

$$Q_3 + 1.5 \times \text{IQR}$$

Interpretation: Values beyond these fences are considered unusual and may indicate:

- Measurement errors
- Data entry mistakes
- Genuine extreme values requiring investigation

4. Comparing Datasets

When comparing two or more datasets:

1. **Compare central tendency:** Which dataset has a higher typical value?
 2. **Compare dispersion:** Which dataset is more consistent/variable?
 3. **Check for outliers:** Are there unusual values affecting the analysis?
 4. **Consider context:** What do the differences mean in the real-world situation?
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Worked Examples

Example 1: Mean, Median, and Mode

Problem: The test scores of 9 students are:

7, 8, 11, 13, 15, 18, 19, 20, 22

Find the mean, median, and mode.

Solution:

Mean:

$$\bar{x} = \frac{7+8+11+13+15+18+19+20+22}{9} = \frac{133}{9} \approx 14.78$$

Median: Data is already ordered. Since

$$n = 9$$

(odd), the median is the middle value (5th position):

$$\text{Median} = 15$$

Mode: No value repeats, so there is **no mode**.

Answer: Mean ≈ 14.78 , Median = 15, No mode

Example 2: Effect of Outliers

Problem: Consider two datasets:

- Dataset A:
5, 6, 7, 8, 9
(mean = 7)

- Dataset B:
5, 6, 7, 8, 50
(same values except last)

Compare the mean and median of both datasets.

Solution:

Dataset A:

- Mean:
$$\frac{5+6+7+8+9}{5} = 7$$

- Median: 7 (middle value)

Dataset B:

- Mean:
$$\frac{5+6+7+8+50}{5} = \frac{76}{5} = 15.2$$

- Median: 7 (middle value unchanged)

Observation: The outlier (50) dramatically increased the mean from 7 to 15.2, but the median remained at 7.

Conclusion: The median is **robust** to outliers, while the mean is **sensitive** to outliers.

Example 3: Quartiles and IQR

Problem: Find Q_1 , Q_2 , Q_3 , and IQR for the dataset:

7, 8, 11, 13, 15, 18, 19, 20, 22

Solution:

Data is already ordered.

$$n = 9$$

(odd).

Q_2 (Median): Middle value (5th position) = **15**

Lower half (below Q_2):

7, 8, 11, 13

$$n = 4$$

(even), so Q_1 = average of 2nd and 3rd values:

$$Q_1 = \frac{8+11}{2} = 9.5$$

Upper half (above Q_2):

18, 19, 20, 22

$$n = 4$$

(even), so Q_3 = average of 2nd and 3rd values:

$$Q_3 = \frac{19+20}{2} = 19.5$$

IQR:

$$\text{IQR} = Q_3 - Q_1 = 19.5 - 9.5 = 10$$

Answer: $Q_1 = 9.5$, $Q_2 = 15$, $Q_3 = 19.5$, $\text{IQR} = 10$

Example 4: Variance and Standard Deviation

Problem: The class sizes of 5 Grade 10 classes are:

43, 45, 46, 41, 40

Find the variance and standard deviation.

Solution:

Step 1: Find the mean:

$$\bar{x} = \frac{43+45+46+41+40}{5} = \frac{215}{5} = 43$$

Step 2: Calculate deviations and squared deviations:

Value (x_i)	Deviation ($x_i - \bar{x}$)	Squared Deviation $(x_i - \bar{x})^2$
43	0	0
45	2	4
46	3	9
41	-2	4
40	-3	9

Sum	0	26
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Step 3: Calculate variance:

$$s^2 = \frac{26}{5} = 5.2$$

Step 4: Calculate standard deviation:

$$s = \sqrt{5.2} \approx 2.28$$

Answer: Variance = 5.2, Standard deviation $\approx$ 2.28

Example 5: Outlier Detection

Problem: A dataset has $Q_1 = 135$ and $Q_3 = 205$. The values 0 and 340 appear in the dataset. Are they outliers?

Solution:

Step 1: Calculate IQR:

$$\text{IQR} = Q_3 - Q_1 = 205 - 135 = 70$$

Step 2: Calculate fences:

- Lower fence:

$$Q_1 - 1.5 \times \text{IQR} = 135 - 1.5(70) = 135 - 105 = 30$$

- Upper fence:

$$Q_3 + 1.5 \times \text{IQR} = 205 + 1.5(70) = 205 + 105 = 310$$

Step 3: Check values:

- $0 < 30 \rightarrow 0$ is an **outlier** (below lower fence)
- $340 > 310 \rightarrow 340$ is an **outlier** (above upper fence)

Answer: Both 0 and 340 are outliers.

Example 6: Comparing Two Datasets

Problem: Two teaching methods were tested on two classes with similar initial English proficiency. After two months, test scores (out of 10) were:

Class A:

2, 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 7, 7, 7, 8, 8, 8, 9, 9, 10

Class B:

3, 3, 4, 5, 5, 5, 6, 6, 6, 7, 7, 7, 7, 8, 8, 9

Which teaching method appears more effective?

Solution:**Class A (**

$n = 20$

):

- Mean:

$$\bar{x}_A = \frac{2+3+4+\dots+10}{20} = \frac{132}{20} = 6.6$$

- Median: Average of 10th and 11th values =

$$\frac{7+7}{2} = 7$$

Class B (

$n = 16$

):

- Mean:

$$\bar{x}_B = \frac{3+3+4+\dots+9}{16} = \frac{100}{16} = 6.25$$

- Median: Average of 8th and 9th values =

$$\frac{6+6}{2} = 6$$

Comparison:

- Class A has higher mean (6.6 vs. 6.25)
- Class A has higher median (7 vs. 6)
- Class A has a wider range (2-10 vs. 3-9)

Conclusion: Class A appears to perform slightly better on average, suggesting Method A may be more effective. However, Class A also shows more variability in performance.

Applications

1. Educational Assessment

Problem: A teacher wants to compare student performance on two exams. Exam 1 scores have mean = 75, standard deviation = 12. Exam 2 scores have mean = 75, standard deviation = 5.

Analysis:

- Both exams have the same average performance
- Exam 2 has smaller standard deviation → students performed more consistently
- Exam 1 has larger standard deviation → greater variation in student performance

Interpretation: Exam 2 may have been clearer or better aligned with instruction, leading to more uniform understanding.

2. Quality Control

Problem: A factory produces bolts with target diameter 10 mm. A sample of 100 bolts has:

- Mean diameter: 10.02 mm
- Standard deviation: 0.15 mm

Analysis:

- Mean is very close to target (only 0.02 mm off)
- Small standard deviation indicates consistent production
- If specifications require diameter within ± 0.5 mm, most bolts meet standards

Interpretation: The production process is well-controlled and meeting quality standards.

3. Income and Wealth Distribution

Problem: In a small town, 9 residents have incomes (in thousands):
25, 30, 32, 35, 38, 40, 42, 45, 300

Should we use mean or median to represent typical income?

Solution:

- Mean:

$$\frac{25+30+\dots+300}{9} = \frac{587}{9} \approx 65.2$$

thousand

- Median: 38 thousand (5th value)

Analysis: The mean (65.2) is much higher than 8 out of 9 incomes due to the outlier (300). The median (38) better represents the typical resident's income.

Conclusion: Use **median** for income data, as it's robust to extreme values and better represents the typical person.

4. Medical Research

Problem: A drug trial measured recovery time (days) for 20 patients:

5, 6, 6, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 10, 10, 11, 12, 13, 15, 42

Identify any outliers and decide whether to include them in the analysis.

Solution:

Calculate quartiles:

- $Q_1 = 7$, $Q_2 = 8.5$, $Q_3 = 10.5$
- $IQR = 10.5 - 7 = 3.5$

Calculate fences:

- Lower:
 $7 - 1.5(3.5) = 1.75$
- Upper:
 $10.5 + 1.5(3.5) = 15.75$

Check: $42 > 15.75 \rightarrow 42$ is an outlier

Interpretation: The patient who took 42 days may have had complications or other factors. Investigate this case separately rather than including it in the main analysis, as it may skew results.

Summary

Key Concepts

Measures of Central Tendency (describe the "center"):

Measure	Formula	Best Used When
Mean	$\bar{x} = \frac{\sum x_i}{n}$	Data is symmetric, no outliers
Median	Middle value	Data is skewed or has outliers
Mode	Most frequent value	Categorical data
Quartiles	Q_1, Q_2, Q_3	Need to understand distribution

Measures of Dispersion (describe the "spread"):

Measure	Formula	Interpretation
Range	Max - Min	Total spread
IQR	$Q_3 - Q_1$	Spread of middle 50%
Variance	$s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$	Average squared deviation
Standard deviation	$s = \sqrt{s^2}$	Typical deviation from mean

Outlier Detection:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

Problem-Solving Strategy

1. **Organize the data:** Sort in ascending order
2. **Calculate central tendency:** Mean, median, mode as appropriate

3. **Calculate dispersion:** Range, IQR, standard deviation
 4. **Check for outliers:** Use the $1.5 \times \text{IQR}$ rule
 5. **Interpret in context:** What do the numbers tell us about the real situation?
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Review Questions

Conceptual Questions

1. Why is the median often preferred over the mean for income data?
2. Can a dataset have more than one mode? Give an example.
3. If every value in a dataset is increased by 10, how do the mean, median, and standard deviation change?
4. If every value in a dataset is multiplied by 2, how do the mean, median, and standard deviation change?
5. Explain why the standard deviation is never negative.

Computational Problems

6. For the dataset:
12, 7, 10, 9, 12, 9, 10, 11, 10, 14
Find: a) Mean, b) Median, c) Mode, d) Q_1 , Q_3 , e) IQR, f) Standard deviation
7. A dataset has mean = 50 and standard deviation = 8. If a value of 70 is added, will the standard deviation increase, decrease, or stay the same? Explain.
8. The five-number summary of a dataset is: Min = 10, $Q_1 = 25$, $Q_2 = 40$, $Q_3 = 55$, Max = 90. Identify any outliers.

Application Problems

9. Two machines produce items with the following quality scores:
 - Machine A: Mean = 85, SD = 3
 - Machine B: Mean = 85, SD = 10Which machine is more consistent? Explain.
0. A class of 25 students has a mean test score of 75. If the teacher adds 5 points to every student's score, what is the new mean?
1. The sodium content (mg per 100g) in 20 cereal brands is measured. The data has $Q_1 = 135$, $Q_3 = 205$. Values of 0 mg and 340 mg appear. Should these be investigated as potential outliers?

🧠 Memory Aids & Tips

Mnemonics

"Mean is MEAN to outliers"

- The mean is affected by (mean to) extreme values, while the median is robust.

"IQR = Inter-Quartile Range = $Q_3 - Q_1$ "

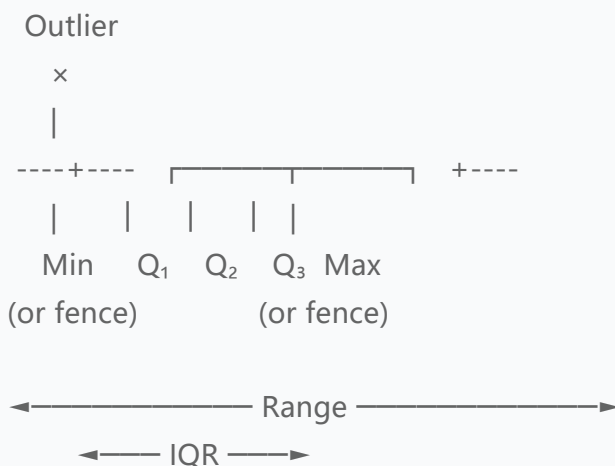
- The range of the middle 50% of data.

"1.5 times IQR = Outlier fence"

- Values beyond
 $Q_1 - 1.5 \times \text{IQR}$
or
 $Q_3 + 1.5 \times \text{IQR}$
are outliers.

Visual Summary: Box Plot

Plain Text



Calculator Tips (Casio fx-580VN X)

Entering data and computing statistics:

1. Enter STAT mode: **MODE** → **3** (STAT)
2. Input data: **60** **=** **90** **=** **120** **=** ...
3. Compute mean: **SHIFT** **1** (STAT) → **4** (Var) → **2** ($\bar{x}$)

(STAT) → 4 (Var) → 2 (
 $\bar{x}$
)

4. Compute standard deviation: SHIFT 1 → 4 → 3 (
 σ_x
)

5. Compute median/quartiles: Use SHIFT 1 → 5 (Data)

Note: For sample standard deviation (
 s
) , use option 4 instead of 3.

Common Pitfalls

1. **Forgetting to order data** before finding median or quartiles
 2. **Confusing variance and standard deviation** (remember:
 $s = \sqrt{s^2}$
)
 3. **Using mean when median is more appropriate** (especially with outliers)
 4. **Misinterpreting standard deviation** (larger SD = more spread, not higher values)
 5. **Forgetting to check for outliers** before computing summary statistics
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Conclusion

Descriptive statistics provides powerful tools for summarizing and understanding data. The key skills are:

1. **Choosing appropriate measures:** Mean vs. median, range vs. IQR
2. **Computing accurately:** Following formulas step-by-step
3. **Interpreting in context:** What do the numbers mean for the real situation?
4. **Identifying unusual values:** Using outlier detection methods
5. **Comparing datasets:** Using both central tendency and dispersion

Connecting to probability: In the previous chapter (Chapter 24), we studied probability as the theoretical foundation for uncertainty. Statistics is the practical counterpart—

using data from the real world to estimate probabilities, test hypotheses, and make informed decisions.

Looking ahead: Advanced statistics courses cover inferential statistics (using samples to make conclusions about populations), hypothesis testing, regression analysis, and more sophisticated modeling techniques.

Practice recommendation: Work through 10-15 problems covering all types of measures, with emphasis on choosing appropriate statistics for different data types and interpreting results in real-world contexts. Use technology (calculator or spreadsheet) to verify calculations for larger datasets.

Congratulations!

You have completed **Chapter 25: Statistics** and the entire **Probability & Statistics** unit!

What you've accomplished:

-  Chapter 23: Combinatorics (counting principles, permutations, arrangements, combinations)
-  Chapter 24: Probability Basics (sample spaces, events, probability rules, independence)
-  Chapter 25: Statistics (descriptive statistics, central tendency, dispersion, outliers)

Total achievement: 25 chapters covering Algebra, Calculus, Geometry, and Probability & Statistics—a comprehensive foundation in high school mathematics!

Next steps: Continue practicing problems, apply these concepts to real-world situations, and explore advanced topics in university-level mathematics, statistics, or data science.

Well done!   